

MODEL CARD

Semantic Similarity-Based Duplicate Question Detection

Sentence-BERT (all-MiniLM-L6-v2) · NLP CA3 · SAIL · BBA (Artificial Intelligence) · AY 2026-27

1 MODEL & TASK		2 DATA & EVALUATION	
Model	Sentence-BERT all-MiniLM-L6-v2 (Sentence Transformers library)	Dataset	Quora Question Pairs (Kaggle): 404,290 labelled pairs; 255,027 non-duplicate (63.08%), 149,263 duplicate (36.92%); 537,933 unique question IDs
Status	Pretrained; not fine-tuned on the Quora data	Split	Unique question IDs split 70 / 15 / 15; a pair is kept only if both IDs fall in the same split. ID sets verified disjoint
Task	Decide whether two English questions express the same meaning or intent. Output: 1 = duplicate, 0 = non-duplicate	Retained	Train 198,177 · Val 8,952 · Test 9,029 = 216,158 pairs (≈53.5%); 188,132 excluded
Embedding	384 dimensions, L2-normalised	Evaluated on	9,029-pair held-out test set (not all 404,290 pairs); duplicate = positive class
Similarity	Cosine similarity between the two question embeddings		
Decision	Duplicate if similarity $\geq$ 0.73 . Threshold chosen on the validation set by maximising F1 (0.10-0.90, step 0.01)		
Purpose	Academic evaluation of duplicate-question candidate detection; not deployed, no automatic merging		

3 TEST PERFORMANCE (9,029 PAIRS, THRESHOLD 0.73)

76.49% Accuracy	61.84% Precision	90.59% Recall	73.51% F1-score
TN 3,961	FP 1,817	FN 306	TP 2,945

4 AUDIT FINDINGS (TWO TEXT SUBGROUPS)

Subgroup	Pairs	Accuracy	Precision	Recall	F1
Low lexical overlap	4,446	85.61%	56.47%	79.28%	65.96%
High lexical overlap	4,583	67.64%	63.46%	94.17%	75.82%
Short question pairs	4,345	73.81%	63.78%	94.16%	76.05%
Long question pairs	4,684	78.97%	58.99%	85.44%	69.79%

Cut-offs are medians: lexical overlap (Jaccard) 0.2727; pair length (average words of the two questions) 10. High overlap: higher recall, lower accuracy. Short: higher precision, recall and F1; long: higher accuracy. Neither group is uniformly better. This is an **association, not causation**.

5 EXPLAINABILITY / FAILURE ANALYSIS (EXAMPLE-LEVEL)

FALSE POSITIVE (1,817 in test set) VERIZON → STRAIGHT TALK vs. STRAIGHT TALK → VERIZON Similarity 0.9950 ; the reversed direction changes the meaning. Similar misses: £1M / <£2,000 vs. more than £1M / <£5,000 (0.9908); Infosys after Nov 21st vs. Nov 7th (0.9904).	FALSE NEGATIVE (306 in test set) “How do I get rid of lethargy?” vs. “...good ways to get rid of weekend lethargy?” Similarity 0.7296 < threshold 0.73 , so it was classified non-duplicate although labelled duplicate (0.7281-0.7299 for the ten highest-scoring false negatives).
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High similarity can miss fine-grained constraints; plausible paraphrases can fall just below the threshold. Not every high-scoring false positive is a clear model failure: some are near-paraphrases labelled non-duplicate, which suggests possible label ambiguity (labels were not manually verified).

6 LIMITATIONS & RESPONSIBLE USE

Limitations - One pretrained Sentence-BERT model; no fine-tuning - One random split, one run; no cross-validation or confidence intervals - Some labels may be ambiguous - Embedding similarity may not preserve direction, numbers, dates, entities or contextual constraints - Questions evaluated were in English; audits used median cut-offs - Results describe the 216,158-pair retained subset	Intended use - Academic NLP experimentation and semantic-similarity research - Duplicate-question candidate identification; decision support for question matching Not intended use - Fully automatic merging or autonomous moderation; human review is required - Treating a similarity score as proof of identical meaning - Use outside English without further evaluation
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